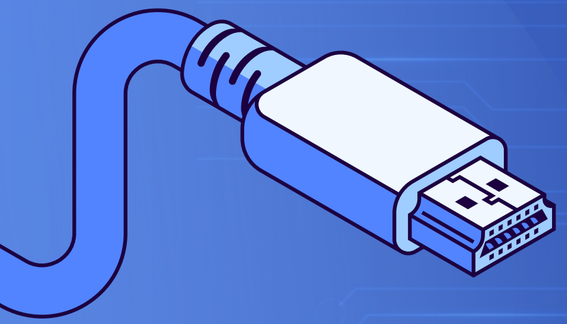
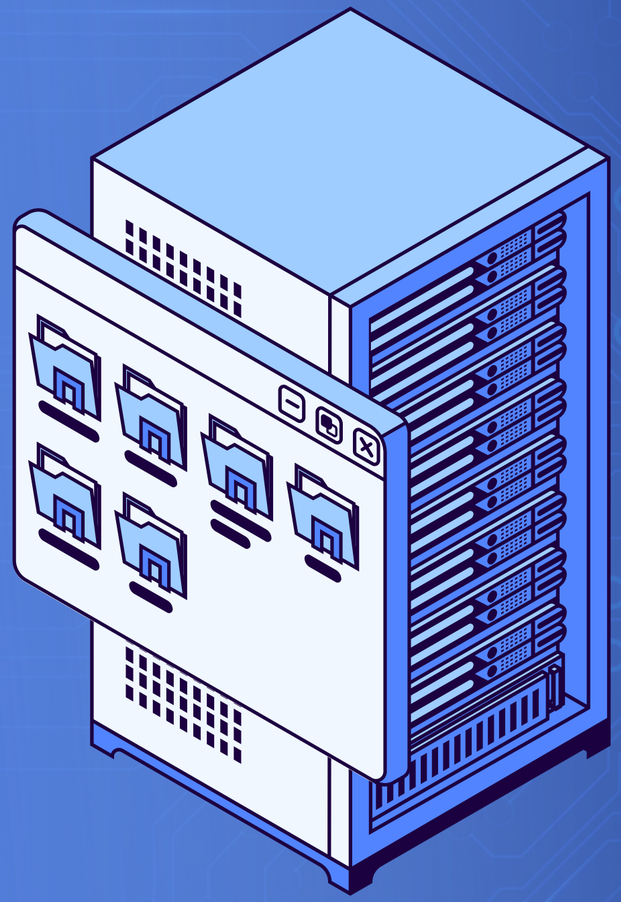


PI PROJECT
PHASE 3





PLAN

Introduction 01.

Kubernetes Architecture 02.

Ansible for Kubernetes 03.

Demonstration 04.

Conclusion 05.

INTRODUCTION

KUBERNETES

"K8S" — THE ORCHESTRATOR

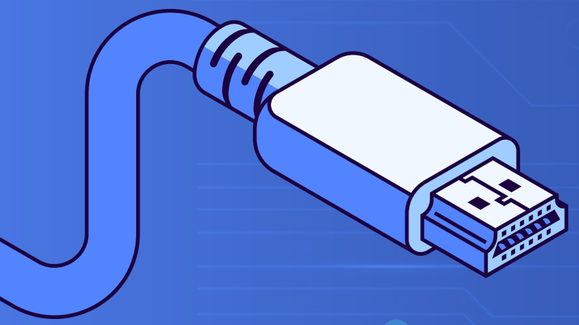
- Open-source platform that automates deploying, scaling and managing containerized applications.
- Originally built by Google, now the industry standard (CNCF).
- Runs the containers, spreads them across machines, and restarts them automatically if they fail.



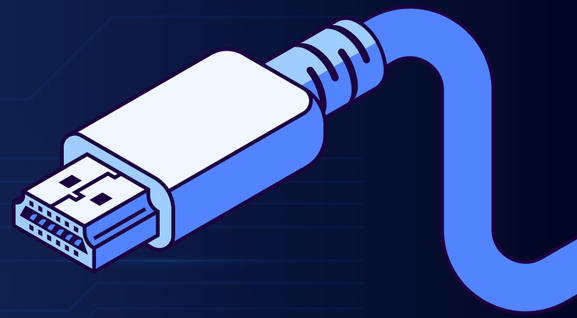
ANSIBLE

THE AUTOMATION ENGINE

- Open-source IT automation tool: installs, configures and deploys servers.
- Agentless — controls machines over plain SSH, nothing to install on targets.
- You describe the desired state in YAML (playbooks); Ansible applies it everywhere, identically.



WHY THEY MATTER IN MODERN DEVOPS ?



The analogy : Ansible builds the stadium (installs and configures the cluster), then Kubernetes is the coach who manages the players (the containers) during the match.

Scalability

More load? K8s adds containers automatically.

High availability

Self-healing: a dead container restarts on its own.

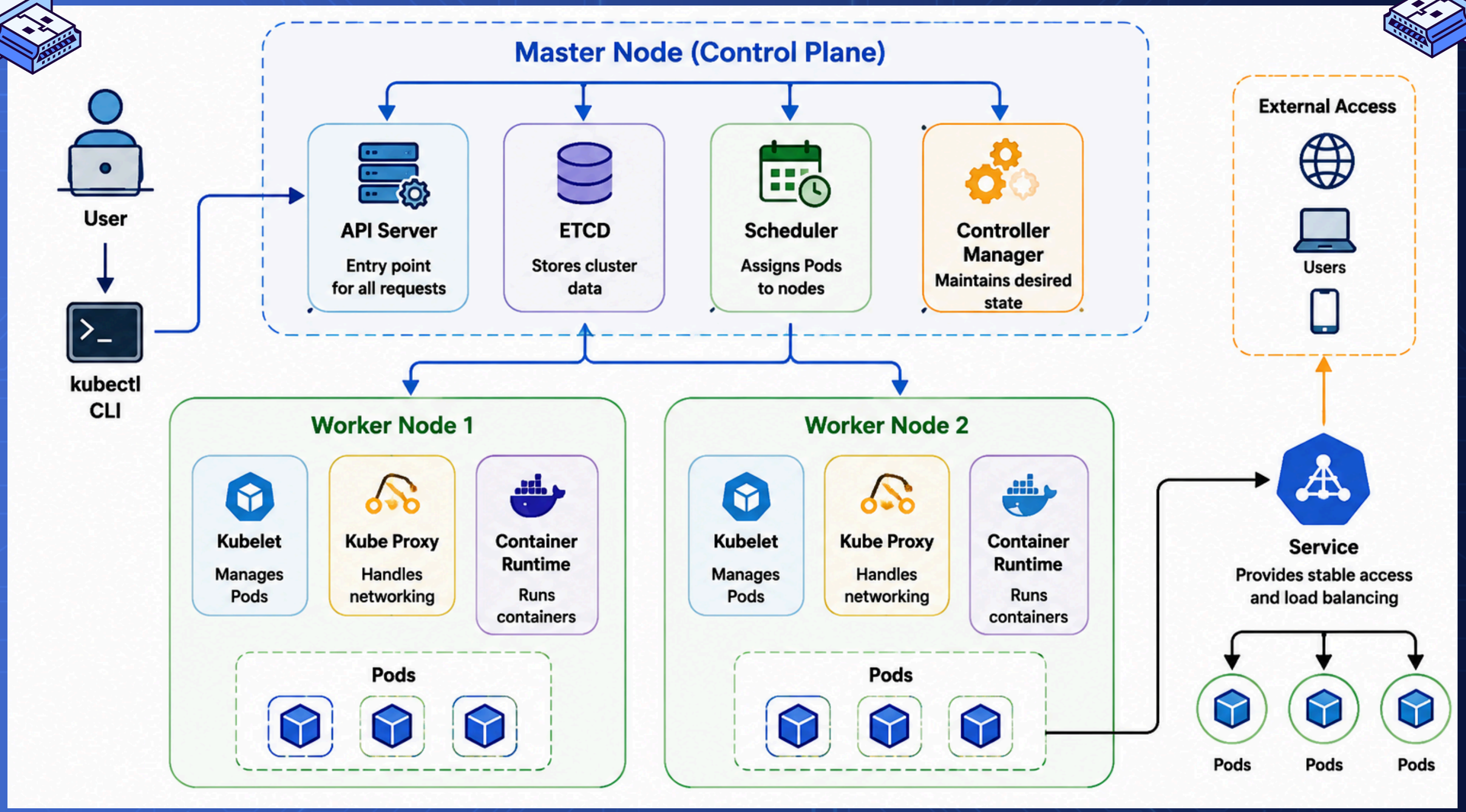
Reproducibility

Ansible replays the same deployment, no human error.

Speed

Cluster setup goes from days to minutes.

KUBERNETES ARCHITECTURE



THE MASTER NODE

THE CLUSTER'S BRAIN

The control plane makes the decisions: what to deploy, where, and it constantly keeps the real state matching the desired state.

API Server

The single entry point. Every command (kubectl) goes through it. It validates and dispatches all requests.

Scheduler

The placer. It decides which Worker Node runs each new Pod, based on available CPU and memory.

Controller Manager

The guardian. It runs control loops and fixes drift: if a Pod is missing, it recreates one.

etcd

The memory. A key-value database that reliably stores the entire state of the cluster.



WORKERS, PODS & SERVICES



Worker Nodes

The machines that do the real work. Each runs the Pods plus kubelet (the agent talking to the master) and kube-proxy (networking).



Pods

The smallest deployable unit. A Pod wraps one or more containers that share network and storage. This is what K8s creates, moves and destroys.



Services

The stable address. Pods come and go (their IP changes); a Service gives a fixed access point and load-balances traffic across them.

HOW ANSIBLE WORKS

4 BUILDING BLOCKS

Inventory

The address book: the list of machines to manage (masters, workers), grouped by role with their IPs.

Playbook

The recipe in YAML: the ordered list of tasks to run on the machines (install, copy, start a service).

Configuration mgmt

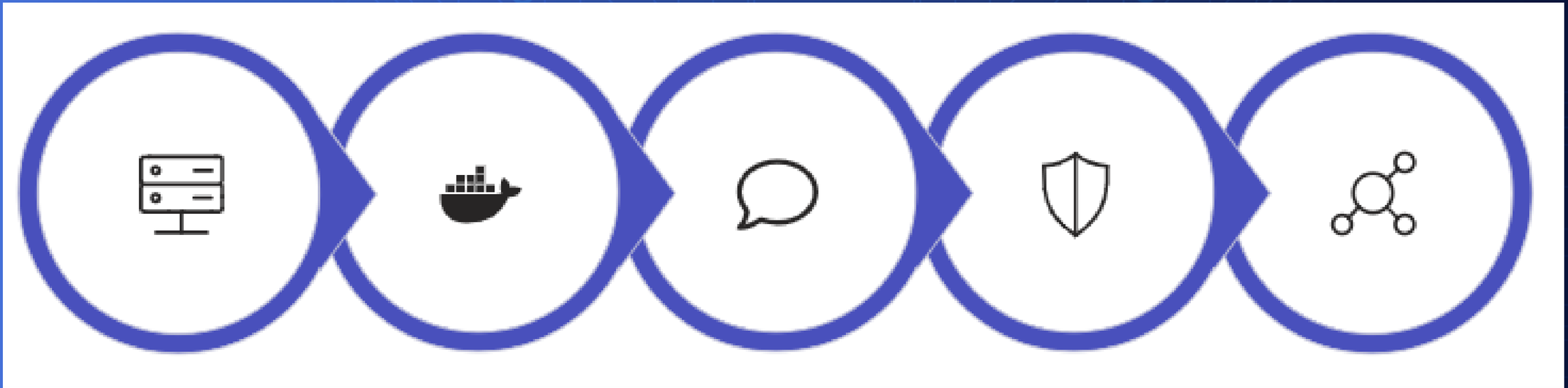
Ansible guarantees each machine ends up in the exact described state — the same everywhere.

Modules & SSH

Agentless: Ansible connects over SSH and runs ready-made modules (apt, copy, service, etc.).

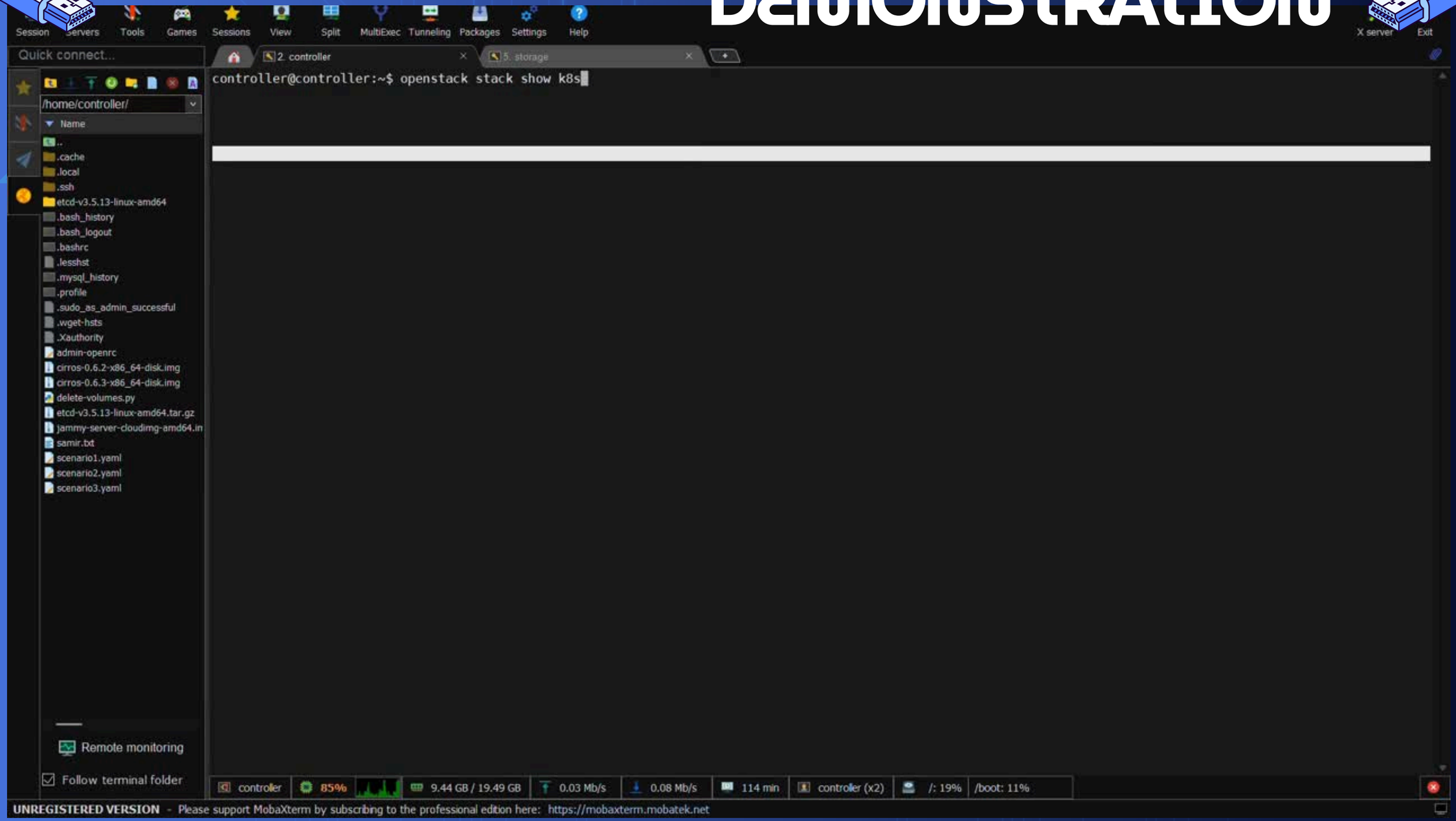
DEPLOYING KUBERNETES WITH ANSIBLE

Each step maps to a distinct Ansible task or role. Running the playbook end-to-end provisions a fully functional, production-ready Kubernetes cluster in minutes — with zero manual intervention.

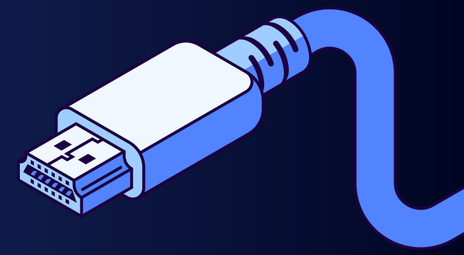


Prepare Hosts	Install Container Runtime	Install Kube Tools	Init Control Plane	Join Worker Nodes

DEMONSTRATION



CONCLUSION



THE KUBERNETES AND ANSIBLE SYNERGY

Ansible prepares and deploys the ground (the cluster). Kubernetes then keeps the applications alive on top.
Together: automated, reproducible, resilient infrastructure.

Automation

No more manual setup: everything is coded and repeatable.

Orchestration

Containers are scheduled, spread and healed automatically.

Resilience

Self-healing and high availability built in.

Scalability

The cluster grows on demand, no rebuild.



THANK YOU.



